# Lecture 15: Sections 4.2 and 4.4

## Relative and Absolute Extrema

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# Relative Maxima and Minima

## Definition (Relative Maximum)

A function $f$ has a **relative maximum** at $c$ if there exists an open interval $(a, b)$ containing $c$ such that $f(x) \leq f(c)$ for all $x \in (a, b)$.

## Definition (Relative Minimum)

A function $f$ has a **relative minimum** at $c$ if there exists an open interval $(a, b)$ containing $c$ such that $f(x) \geq f(c)$ for all $x \in (a, b)$.

## Critical Point

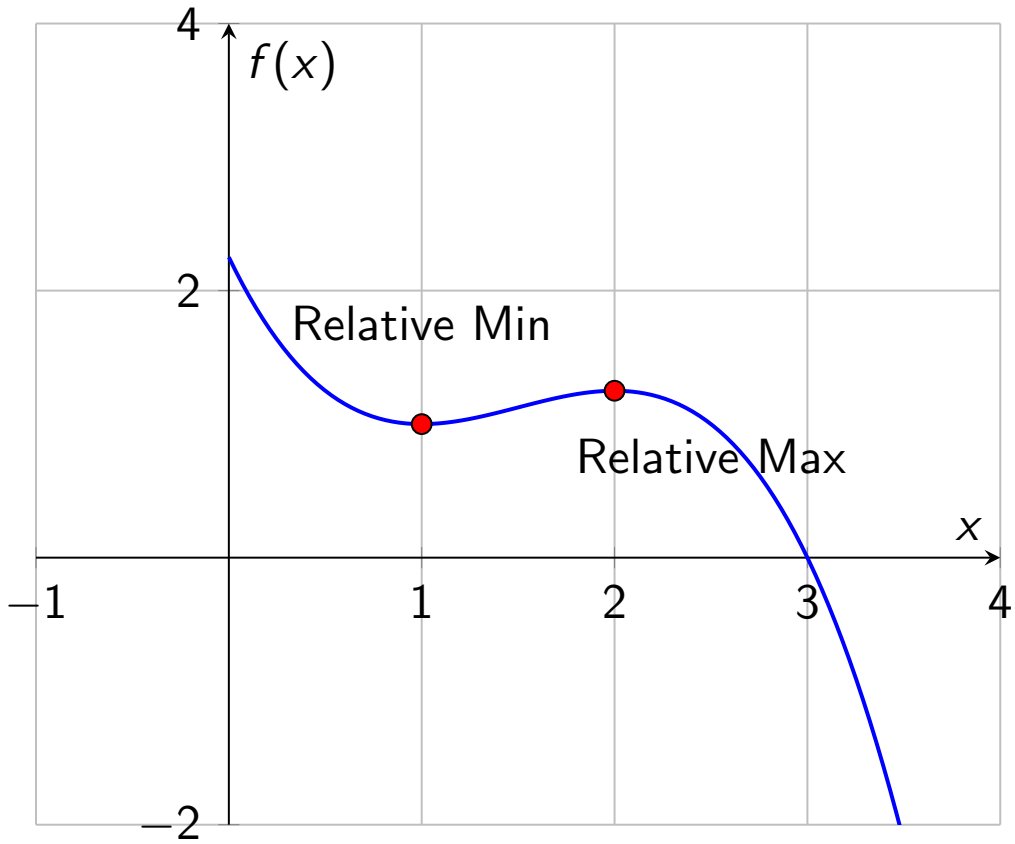
A point $c$ in the domain of $f$ is called a **critical point** if either:

- $f'(c) = 0$, or
- $f'(c)$ does not exist

## Theorem

*If $f$ has a relative extremum at $c$, then $c$ is a critical point of $f$.*

# Visualizing Relative Extrema



# First Derivative Test

## Theorem (First Derivative Test)

*Let $c$ be a critical point of a continuous function $f$.*

- *If $f'$ changes from* **positive to negative** *at $c$, then $f$ has a* **relative maximum** *at $c$*
- *If $f'$ changes from* **negative to positive** *at $c$, then $f$ has a* **relative minimum** *at $c$*
- *If $f'$ does* **not change sign** *at $c$, then $f$ has* **no relative extremum** *at $c$*

## Procedure

1. Find all critical points
2. Create sign chart for $f'(x)$
3. Analyze sign changes around each critical point

# Example 1: First Derivative Test

Find relative extrema of $f(x) = x^3 - 3x^2 - 9x + 5$.

**Solution:**

1. $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$
2. Critical points: $x = -1, 3$
3. Sign analysis:
   - $(-\infty, -1)$: $f'(x) > 0$ (increasing)
   - $(-1, 3)$: $f'(x) < 0$ (decreasing)
   - $(3, \infty)$: $f'(x) > 0$ (increasing)
4. $f(-1) = (-1)^3 - 3(1) - 9(-1) + 5 = 10$
5. $f(3) = 27 - 27 - 27 + 5 = -22$

**Answer:** Relative maximum at $(-1, 10)$, Relative minimum at $(3, -22)$

# Second Derivative Test

## Theorem (Second Derivative Test)

*Let $f$ be twice differentiable at $c$ with $f'(c) = 0$.*

- *If $f''(c) > 0$, then $f$ has a* **relative minimum** *at $c$*
- *If $f''(c) < 0$, then $f$ has a* **relative maximum** *at $c$*
- *If $f''(c) = 0$, the test is* **inconclusive**

## Advantages & Limitations

- **Faster** than first derivative test when applicable
- **Doesn't work** when $f''(c) = 0$ or $f''(c)$ doesn't exist
- Only applies to points where $f'(c) = 0$ (not where derivative DNE)

# Example 2: Second Derivative Test

Find relative extrema of $f(x) = x^4 - 4x^3 + 6x^2$.

**Solution:**

1. $f'(x) = 4x^3 - 12x^2 + 12x = 4x(x^2 - 3x + 3)$
2. Critical point: $x = 0$ (only real solution)
3. $f''(x) = 12x^2 - 24x + 12$
4. $f''(0) = 12 > 0$

**Answer:** Relative minimum at $(0, 0)$

## Note

The quadratic $x^2 - 3x + 3$ has discriminant $9 - 12 = -3 < 0$, so no real roots.

# Geometric Implications of Multiplicity

For polynomial $f(x) = a(x - r_1)^{m_1}(x - r_2)^{m_2} \cdots (x - r_k)^{m_k}$:

## Theorem (Multiplicity and Graph Behavior)

*At a zero $x = r$ with multiplicity $m$:*

- *If $m$ is* **odd***: Graph* **crosses** *the x-axis*
- *If $m$ is* **even***: Graph* **touches** *the x-axis and turns around*
- *If $m = 1$: Graph crosses* **straight through** *(linear crossing)*
- *If $m > 1$: Graph is* **flatter** *near the zero*

## At Critical Points

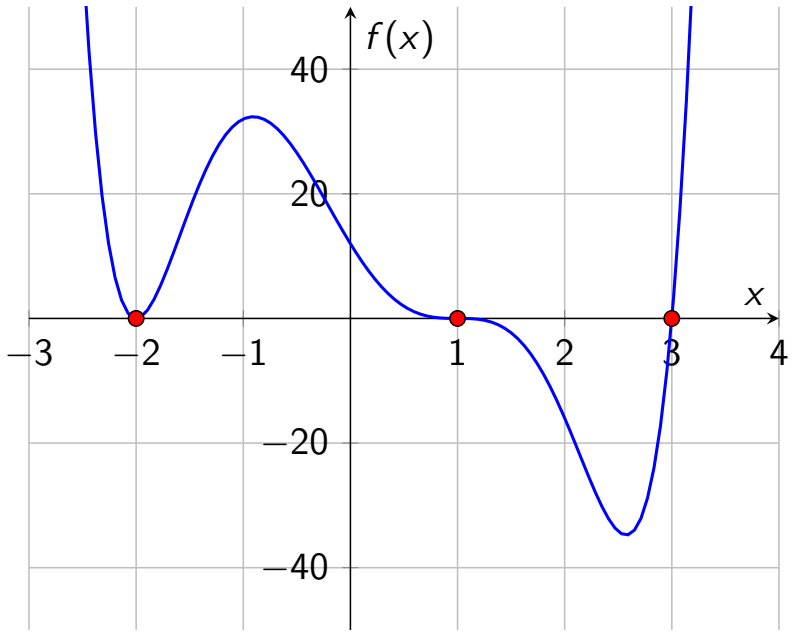
- Odd multiplicity $> 1$ in derivative: Inflection point
- Even multiplicity in derivative: Extremum point

# Example 3: Multiplicity Analysis

Analyze $f(x) = (x-1)^3(x+2)^2(x-3)$.

**Solution:**

- Zero at $x = 1$, multiplicity 3 (odd): **Crosses** with flattening
- Zero at $x = -2$, multiplicity 2 (even): **Touches** and turns
- Zero at $x = 3$, multiplicity 1 (odd): **Crosses** straight through
- Degree: $3 + 2 + 1 = 6$ (even), positive leading coefficient
- End behavior: $\lim_{x \to \pm\infty} f(x) = +\infty$



# Complete Polynomial Analysis Procedure

1. Find all **zeros** and their **multiplicities**
2. Determine **end behavior** using leading term
3. Find **y-intercept**: $f(0)$
4. Find **critical points**: Solve $f'(x) = 0$
5. Use **first or second derivative test** to classify extrema
6. Find **inflection points**: Solve $f''(x) = 0$
7. Determine **concavity** intervals
8. Sketch the graph

# Example 4: Complete Polynomial Analysis

Analyze $f(x) = x^4 - 4x^3 + 4x^2$.

**Solution:**

1. Zeros: $f(x) = x^2(x-2)^2$, zeros at $x = 0$ (m=2), $x = 2$ (m=2)
2. End behavior: Degree 4, positive $\Rightarrow \uparrow\uparrow$
3. y-intercept: $f(0) = 0$
4. $f'(x) = 4x^3 - 12x^2 + 8x = 4x(x-1)(x-2)$
5. Critical points: $x = 0, 1, 2$
6. $f''(x) = 12x^2 - 24x + 8 = 4(3x^2 - 6x + 2)$
7. Inflection points: $x = 1 \pm \frac{1}{\sqrt{3}}$

**Extrema:**

- $x = 0$: $f''(0) = 8 > 0 \Rightarrow$ Relative min
- $x = 1$: $f''(1) = -4 < 0 \Rightarrow$ Relative max
- $x = 2$: $f''(2) = 8 > 0 \Rightarrow$ Relative min

# Absolute Maxima and Minima

## Definition (Absolute Maximum)

A function $f$ has an **absolute maximum** at $c$ if $f(c) \geq f(x)$ for all $x$ in the domain of $f$.

## Definition (Absolute Minimum)

A function $f$ has an **absolute minimum** at $c$ if $f(c) \leq f(x)$ for all $x$ in the domain of $f$.

## Key Differences from Relative Extrema

- **Relative**: Only compared to nearby points
- **Absolute**: Compared to **all** points in domain
- A function can have multiple relative extrema but only one absolute max/min

# The Extreme Value Theorem

## Theorem (Extreme Value Theorem)

*If $f$ is* **continuous** *on a* **closed interval** $[a, b]$*, then $f$ attains both an* **absolute maximum** *and an* **absolute minimum** *on* $[a, b]$*.*

## Where Absolute Extrema Can Occur

On $[a, b]$, absolute extrema can occur at:

1. **Critical points** inside $(a, b)$
2. **Endpoints** $a$ and $b$

## Important

The Extreme Value Theorem **only guarantees** existence of absolute extrema for **continuous** functions on **closed** intervals.

# Procedure: Absolute Extrema on Closed Interval

To find absolute extrema of continuous $f$ on $[a, b]$:

1. Find all **critical points** in $(a, b)$
2. Evaluate $f$ at all **critical points**
3. Evaluate $f$ at **endpoints** $a$ and $b$
4. The **largest** value is absolute maximum
5. The **smallest** value is absolute minimum

# Example 5: Absolute Extrema on Closed Interval

Find absolute extrema of $f(x) = x^3 - 3x^2 + 1$ on $[-1, 4]$.

**Solution:**

1. $f'(x) = 3x^2 - 6x = 3x(x - 2)$
2. Critical points: $x = 0, 2$ (both in $[-1, 4]$)
3. Evaluate:
   - $f(-1) = -1 - 3 + 1 = -3$
   - $f(0) = 1$
   - $f(2) = 8 - 12 + 1 = -3$
   - $f(4) = 64 - 48 + 1 = 17$

**Answer:**

- Absolute maximum: 17 at $x = 4$
- Absolute minimum: $-3$ at $x = -1$ and $x = 2$

# Absolute Extrema on Infinite Intervals

For intervals like $(-\infty, \infty)$, $[a, \infty)$, $(-\infty, b]$:

## Procedure

1. Find all critical points
2. Analyze **end behavior** using limits
3. If function $\to \pm\infty$, no absolute max/min in that direction
4. Compare function values at critical points

## Example

$f(x) = x^2$ on $(-\infty, \infty)$:

- Absolute min at $x = 0$, no absolute max

# Example 6: Absolute Extrema on Infinite Interval

Find absolute extrema of $f(x) = \dfrac{x}{x^2+1}$ on $(-\infty, \infty)$.

**Solution:**

1. $f'(x) = \dfrac{(1)(x^2+1) - x(2x)}{(x^2+1)^2} = \dfrac{1-x^2}{(x^2+1)^2}$
2. Critical points: $x = -1, 1$
3. End behavior: $\lim_{x\to\pm\infty} f(x) = 0$
4. Evaluate:
   - $f(-1) = -\frac{1}{2}$
   - $f(1) = \frac{1}{2}$

**Answer:**

- Absolute maximum: $\frac{1}{2}$ at $x = 1$
- Absolute minimum: $-\frac{1}{2}$ at $x = -1$

# Absolute Extrema on Open Intervals

For intervals like $(a, b)$, $(a, \infty)$, $(-\infty, b)$:

## Key Issue

Extreme Value Theorem **does not apply** - absolute extrema may not exist!

## Procedure

1. Find all critical points in $(a, b)$
2. Evaluate $f$ at critical points
3. Analyze **limits at endpoints**
4. If limits suggest higher/lower values, no absolute max/min

# Example 7: Absolute Extrema on Open Interval

Find absolute extrema of $f(x) = x^2$ on $(0, 2)$.

**Solution:**

1. $f'(x) = 2x$, critical point: $x = 0$ (not in $(0, 2)$)
2. No critical points in interval
3. End behavior:
   - $\lim_{x \to 0^+} f(x) = 0$
   - $\lim_{x \to 2^-} f(x) = 4$
4. Function takes values arbitrarily close to 0 and 4

**Answer:**

- No absolute minimum (can get arbitrarily close to 0)
- No absolute maximum (can get arbitrarily close to 4)

# Functions with One Relative Extremum

## Theorem

*If a continuous function $f$ has* **exactly one relative extremum** *on an interval $I$ and it is a relative maximum (minimum), then it is also the* **absolute maximum (minimum)** *on $I$.*

## Common Examples

- Quadratic functions: $f(x) = ax^2 + bx + c$
- Functions of form $f(x) = x^n$ on $(0, \infty)$
- Many exponential and logarithmic functions

# Example 8: One Relative Extremum

Find absolute extrema of $f(x) = x^2 e^{-x}$ on $[0, \infty)$.

**Solution:**

1. $f'(x) = 2xe^{-x} - x^2 e^{-x} = xe^{-x}(2 - x)$
2. Critical points: $x = 0, 2$
3. $f''(x) = e^{-x}(x^2 - 4x + 2)$
4. $f''(0) = 2 > 0 \Rightarrow$ Relative min at $x = 0$
5. $f''(2) = -2e^{-2} < 0 \Rightarrow$ Relative max at $x = 2$
6. Only one relative maximum in $(0, \infty)$
7. $\lim_{x \to \infty} f(x) = 0$

**Answer:**

- Absolute maximum: $f(2) = 4e^{-2}$ at $x = 2$
- Absolute minimum: $f(0) = 0$ at $x = 0$

# Practice Problems - Relative Extrema

1. Find all relative extrema of $f(x) = x^3 - 12x + 5$
2. Locate relative extrema of $g(x) = \dfrac{x^2}{x-1}$
3. Find critical points and classify for $h(x) = x^{4/3} - 4x^{1/3}$
4. Determine relative extrema of $k(x) = \sin x + \cos x$ on $[0, 2\pi]$
5. Find relative max/min of $m(x) = xe^{-x^2}$
6. Classify extrema of $n(x) = x^2(x-3)^2$
7. Find relative extrema of $p(x) = \dfrac{\ln x}{x}$
8. Locate and classify extrema of $q(x) = x + \dfrac{1}{x}$

# Practice Problems - Absolute Extrema

1. Find absolute extrema of $f(x) = x^3 - 3x + 1$ on $[0, 3]$
2. Find absolute max/min of $g(x) = \dfrac{x}{x^2 + 4}$ on $[-2, 2]$
3. Determine absolute extrema of $h(x) = x\sqrt{4 - x^2}$ on $[-2, 2]$
4. Find absolute max/min of $k(x) = e^{-x} \sin x$ on $[0, \pi]$
5. Locate absolute extrema of $m(x) = |x^2 - 4|$ on $[-3, 3]$
6. Find absolute extrema of $n(x) = x + 2\cos x$ on $[-\pi, \pi]$
7. Determine absolute max/min of $p(x) = \dfrac{x^2 - 1}{x^2 + 1}$ on $(-\infty, \infty)$
8. Find absolute extrema of $q(x) = x^4 - 8x^2$ on $(-\infty, \infty)$

# Practice Problems - Applications & Analysis

1. A rectangular field is bounded on one side by a river. With 400m of fencing, find dimensions that maximize area.
2. Find point on parabola $y = x^2$ closest to $(0, 3)$
3. Analyze $f(x) = x^4 - 8x^3 + 18x^2$ completely
4. A box with square base and open top must have volume 32,000 $cm^3$. Find dimensions that minimize surface area.
5. Analyze $g(x) = \dfrac{x^3}{x^2 - 4}$ completely
6. Find dimensions of cylinder with volume $54\pi$ that minimizes surface area
7. Analyze $h(x) = e^{-x^2/2}$ completely
8. A right triangle has hypotenuse 10. Find maximum area.

# Summary of Key Concepts

- **Relative extrema** occur at critical points where derivative changes sign
- **Absolute extrema** consider entire domain
- **Extreme Value Theorem** guarantees existence on closed intervals
- **First derivative test** analyzes sign changes
- **Second derivative test** uses concavity (when applicable)
- **Multiplicity** affects graph behavior at zeros
- Different procedures for different interval types

**Remember:** Always check endpoints for absolute extrema on closed intervals!